

Using MySQL and Pivot Tables for Data Analysis

Data analysis requires the extraction, cleansing, transformation, sorting and the modelling of data in order to derive useful results from it—to make useful deductions and support decision making. Pivot tables are important in the field of data analysis because they help in summarising data quickly and efficiently.



A pivot table is a data organisation tool used frequently by data analysts to get summary reports of data in different formats. It is widely used to evaluate basic descriptive statistics of data. These include sorting, reorganising, grouping, counting, summarising and averaging of the recorded data. These statistics are used for further higher level data processing like pattern analysis or decision making. A pivot table allows data scientists to transform columns into rows and vice versa. It also allows data grouping by its constituent attribute fields. It provides a better solution for summarising a large volume of data and is widely used in Microsoft Excel. The pivot table concept can be implemented using SQL queries too. This technique is discussed here with a few examples on some common data sets. A pivot table requires a CASE statement to switch rows to columns. Let's briefly explore CASE statements.

MySQL CASE expression

MySQL CASE expression is a control flow structure that allows *if-else* logic in a query. It can be used with SELECT, WHERE and ORDER BY clauses. The CASE expression has

two forms: simple CASE and searched CASE. Simple CASE is used in the stored procedure and search CASE is used in the SELECT clause for queries.

A simple CASE syntax is as follows:

```
CASE value
WHEN value1 THEN result1
WHEN value2 THEN result2
...
[ELSE else_result]
```

A search CASE syntax is as follows:

```
CASE
WHEN expression1 THEN result1
WHEN expression2 THEN result2
...
[ELSE else_result]
END
```

A simple CASE statement acts on the basis of the 'value' parameter, whereas the search CASE statement acts on the